Improving Cantonese translation in existing multilingual translation models

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Abstract

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1 Related works

1.1 MADLAD

2 Introduction

Although Cantonese has more than 85 million speakers, it is very underrepresented in machine learning and translation, with many Cantonese tasks simply being handled by models trained mostly or exclusively on Mandarin. In this project, we explore how both unsupervised and supervised learning can be applied to improve the quality of translation tasks. For this, we use the professionally translated data in the SMOL[1] and GATITOS[3] datasets and the unlabeled data in the dataset collected by Dare et al.[2]. We use the unlabeled data to train Cantonese token embeddings and then transform these embeddings into the embedding space of a pre-trained English-Mandarin model, MADLAD[?], using the token-level translations in GATITOS. This TODO: Results of this. We then finetune MADLAD on the SMOL dataset and evaluate the performance of the finetuned model on a held-out test set from SMOL. TODO: Results of this.

3 Datasets

3.1 SMOL

SMOL[1] (Set of Maximal Leverage) is a high-quality translation corpus from English to more than a hundred low-resource languages. SMOL includes SMOLSent, SMOLDoc and Gatitos[3] which represent sentence-level, document-level and token-level translations respectively.

3.2 Monolingual Cantonese corpus by M. Dare, et. al

In their work on unsupervised Mandarin-Cantonese machine translation, Dare et. al.[2] collected a large corpus of monolingual Cantonese sentences from various online sources. In total 912258

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lines were collected by scraping and cleaning data from Wikipedia, Instagram, and a few other sites including a few corpusses compiled for previous academic work. Common for most sources that form the corpus is that the content is mostly

4 Experiments

4.1 Embedding transform

If we were to simply train a model on Gatitos token-level data it might get slightly better at some characters in the small Gatitos dataset, but we would not be able to generalize the model to become better at all Cantonese vocabulary.

Instead we propose the following experiment, we train embeddings on a large corpus of unlabeled Cantonese data, hoping to get embeddings that better represent semantic similarity between Cantonese tokens than the embeddings of MADLAD do. We then perform an embedding transform from our Cantonese embeddings to MADLAD's Chinese embedding space using token-level translations from Gatitos. Specifically, we train a transformation matrix W such that for a specific Cantonese-Chinese token embedding pair $E_{canto}, E_{mandarin}$ the embedding transformation $E_{canto}W$ approximates $E_{mandarin}$. This training was simply done optimizing the loss function $||E_{canto}W - E_{mandarin}||$ with the ADAM-algorithm[4].

For training the Cantonese embeddings we use the Monolingual Cantonese corpus by M. Dare, et. al.

4.2 Smol finetuning

5 Results

5.1 Embedding transform

5.2 Smol finetuning

6 Discussion

References

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